INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL

Department of Mathematics

Engineering Mathematics – I (CSE-1001)

Mini-Test, 1 October 2026 — Solutions and Marking Scheme

B.Tech Semester I · CS (AI), Section 6 · Maximum Marks 20

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1 (A)(a) — Differentiable except at two points [1]

Give an example of a function $f : \mathbb{R} \to \mathbb{R}$ which is differentiable at every point of $\mathbb{R}$ except at $x = 2026$ and $x = -2026$.

A valid example is

$$f(x) = \left|x^2 - 2026^2\right|.$$

Justification. Write $u(x) = x^2 - 2026^2$, a polynomial, so u is differentiable on all of $\mathbb{R}$. The absolute value $|u|$ is differentiable wherever $u \neq 0$, and may fail to be differentiable only where $u = 0$, that is at $x = \pm 2026$. At those two points it does fail: at $x = 2026$,

$$f'(2026^+) = \lim_{h \to 0^+} \frac{|(2026+h)^2 - 2026^2|}{h} = 2(2026) = 4052, \qquad f'(2026^-) = -4052,$$

and the one-sided derivatives are unequal, so f is not differentiable there. The same computation at $x = -2026$ gives ∓ 4052. Hence f is differentiable precisely on $\mathbb{R} \setminus \{\pm 2026\}$.

Other acceptable answers. $f(x) = |x - 2026| + |x + 2026|$; $f(x) = |x - 2026| \cdot |x + 2026|$; or any piecewise definition whose only two points of failure are $x = \pm 2026$.

1 (A)(b) — Differentiability of $x^2 \sin(1/x)$ at the origin [1]

$f(x) = x^2 \sin(1/x)$ for $x \neq 0$, $f(0) = 0$. Show that f is differentiable at $x = 0$.

By definition,

$$f'(0) \;=\; \lim_{h \to 0} \frac{f(h) - f(0)}{h} \;=\; \lim_{h \to 0} \frac{h^2 \sin(1/h) - 0}{h} \;=\; \lim_{h \to 0} h \sin\!\left(\frac{1}{h}\right).$$

Since $|\sin(1/h)| \leq 1$ for every $h \neq 0$,

$$0 \;\leq\; \left|h \sin\!\left(\tfrac{1}{h}\right)\right| \;\leq\; |h|,$$

and $|h| \to 0$ as $h \to 0$. By the Sandwich Theorem the limit exists and equals 0. Hence f is differentiable at $x = 0$ with

$$\boxed{f'(0) = 0}.$$

1 (B)(a) — Differentiability implies continuity [2]

Prove that if f is differentiable at $x = a$ then f is continuous at $x = a$. Is the converse true? Justify with an example.

Proof. Suppose $f'(a)$ exists. For $x \neq a$ write the identity

$$f(x) - f(a) \;=\; \frac{f(x) - f(a)}{x - a} \cdot (x - a).$$

Letting $x \to a$ and using the algebra of limits, the first factor tends to $f'(a)$ (which exists and is finite by hypothesis) and the second tends to 0. Therefore

$$\lim_{x \to a} \big(f(x) - f(a)\big) \;=\; f'(a) \cdot 0 \;=\; 0, \qquad \text{that is} \qquad \lim_{x \to a} f(x) = f(a),$$

which is precisely continuity of f at a. ■

The converse is false. Take $f(x) = |x|$ at $a = 0$.

Continuity: $\lim_{x \to 0} |x| = 0 = f(0)$, so f is continuous at 0.

Non-differentiability:

$$\lim_{h \to 0^+} \frac{|h| - 0}{h} = \lim_{h \to 0^+} \frac{h}{h} = 1, \qquad \lim_{h \to 0^-} \frac{|h| - 0}{h} = \lim_{h \to 0^-} \frac{-h}{h} = -1.$$

The one-sided limits differ, so $f'(0)$ does not exist. Hence continuity does not imply differentiability.

1 (B)(b) — Unequal finite one-sided derivatives [1]

The one-sided limits of the difference quotient at a both exist finitely but are unequal. Is f necessarily continuous at a?

Yes — f is necessarily continuous at a.

Write L^- and L^+ for the two one-sided limits, both finite by hypothesis. For $x < a$,

$$f(x) - f(a) = \frac{f(x) - f(a)}{x - a} \cdot (x - a) \;\xrightarrow[x \to a^-]{}\; L^- \cdot 0 = 0,$$

so $\lim_{x \to a^-} f(x) = f(a)$. The identical computation on the right gives $\lim_{x \to a^+} f(x) = f(a)$. Both one-sided limits of f exist and equal $f(a)$, hence $\lim_{x \to a} f(x) = f(a)$ and f is continuous at a.

Remark. What the hypothesis $L^- \neq L^+$ rules out is *differentiability*, not continuity. Continuity needs only that each one-sided limit be *finite*; their agreeing is an extra condition, and it is exactly the extra condition that differentiability requires. The function $f(x) = |x|$ at $a = 0$ realises this situation with $L^- = -1$, $L^+ = +1$.

2 (A)(a) — Four sequences with prescribed behaviour [2]

Give, with justification, one example of a sequence which is (i) bounded, not monotone, convergent; (ii) bounded, not monotone, divergent; (iii) monotone and unbounded; (iv) neither bounded nor monotone.

	Example	Justification
(i)	$a_n = \dfrac{(-1)^n}{n}$	$\lvert a_n\rvert \le 1$, so bounded. The terms alternate in sign ($a_1 = -1 < a_2 = \frac{1}{2} > a_3 = -\frac{1}{3}$), so not monotone. $\lvert a_n\rvert = 1/n \to 0$, so $a_n \to 0$: convergent.
(ii)	$a_n = (-1)^n$	$\lvert a_n\rvert = 1$, so bounded. Alternates, so not monotone. The subsequences $a_{2n} = 1$ and $a_{2n+1} = -1$ have different limits, so no limit exists: divergent.
(iii)	$a_n = n$	Strictly increasing, hence monotone. Unbounded, since for any M there is $n > M$. (It diverges to $+\infty$.)
(iv)	$a_n = (-1)^n n$	$\lvert a_n\rvert = n \to \infty$, so unbounded. The terms $-1, 2, -3, 4, \ldots$ neither increase nor decrease, so not monotone.

2 (A)(b) — Convergence of $a_n = n!/n^n$ [1]

Examine the convergence of the sequence $a_n = \dfrac{n!}{n^n}$.

Method 1 (direct, Sandwich). Write the quotient out:

$$a_n \;=\; \frac{1 \cdot 2 \cdot 3 \cdots n}{n \cdot n \cdot n \cdots n} \;=\; \frac{1}{n} \cdot \frac{2}{n} \cdot \frac{3}{n} \cdots \frac{n}{n}.$$

Every factor after the first is at most 1, so

$$0 \;<\; a_n \;\le\; \frac{1}{n} \qquad (n \ge 1).$$

Since $1/n \to 0$, the Sandwich Theorem gives $a_n \to 0$: the sequence **converges, with limit** 0.

Method 2 (ratio test for sequences).

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!} = \frac{n^n}{(n+1)^n} = \frac{1}{\left(1 + \frac{1}{n}\right)^n} \;\longrightarrow\; \frac{1}{e} \approx 0.3679 < 1,$$

so by the ratio test for sequences $a_n \to 0$.

2 (B) — Limit of a sum of n terms [2]

Evaluate $\displaystyle\lim_{n\to\infty} \left(\frac{1}{\sqrt{n^2+1}} + \frac{1}{\sqrt{n^2+2}} + \cdots + \frac{1}{\sqrt{n^2+n}} \right)$.

Write $a_n = \displaystyle\sum_{k=1}^{n} \frac{1}{\sqrt{n^2+k}}$. There are exactly n terms, and since $\sqrt{n^2+k}$ increases with k, the largest term is the one with $k = 1$ and the smallest the one with $k = n$. Replacing every term by the largest, and then by the smallest, gives

$$\underbrace{\frac{n}{\sqrt{n^2+n}}}_{\text{all terms} = \text{smallest}} \;\le\; a_n \;\le\; \underbrace{\frac{n}{\sqrt{n^2+1}}}_{\text{all terms} = \text{largest}} .$$

Dividing numerator and denominator of each bound by n,

$$\frac{1}{\sqrt{1+\dfrac{1}{n}}} \;\le\; a_n \;\le\; \frac{1}{\sqrt{1+\dfrac{1}{n^2}}}.$$

Both outer expressions tend to 1 as $n \to \infty$. By the **Sandwich (Squeeze) Theorem**,

$$\boxed{\lim_{n\to\infty} a_n = 1}.$$

Numerical check (for the marker). At $n = 10$: $0.95346 \le 0.97386 \le 0.99504$. At $n = 10^6$: $0.9999995 \le 0.9999998 \le 1.0000000$.

3 (A) — Convergence of three series [3]

(i) $\displaystyle\sum_{n=1}^{\infty} \frac{n!}{(n^n)^2} = \sum_{n=1}^{\infty} \frac{n!}{n^{2n}}$ [1]

Comparison (quickest). For $n \ge 2$,

$$\frac{n!}{n^{2n}} \;\le\; \frac{n^n}{n^{2n}} \;=\; \frac{1}{n^n} \;\le\; \frac{1}{2^n},$$

using $n! \le n^n$ and $n \ge 2$. Since $\sum 2^{-n}$ is a convergent geometric series ($r = \frac{1}{2} < 1$), the given series **converges** by the comparison test.

Ratio test (alternative).

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)!}{(n+1)^{2n+2}} \cdot \frac{n^{2n}}{n!} = \frac{n^{2n}}{(n+1)^{2n+1}} = \left(\frac{n}{n+1}\right)^{2n} \cdot \frac{1}{n+1} \;\longrightarrow\; e^{-2} \cdot 0 = 0 < 1,$$

so the series converges. (Numerically the ratio is 0.0269 at $n = 5$ and 0.00226 at $n = 60$.)

(ii) $\displaystyle\sum_{n=1}^{\infty} \frac{n^{2026}}{2026^{\,n}}$ [1]

By D'Alembert's ratio test,

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)^{2026}}{2026^{\,n+1}} \cdot \frac{2026^{\,n}}{n^{2026}} = \frac{1}{2026}\left(1+\frac{1}{n}\right)^{2026} \;\longrightarrow\; \frac{1}{2026} \;<\; 1.$$

Hence the series **converges**.

(iii) $\dfrac{5}{2} - \dfrac{7}{4} + \dfrac{9}{6} - \dfrac{11}{8} + \cdots$ [1]

The general term is

$$u_n = (-1)^{n+1}\,\frac{2n+3}{2n}, \qquad \text{so} \qquad |u_n| = \frac{2n+3}{2n} = 1 + \frac{3}{2n} \;\longrightarrow\; 1.$$

Since $u_n \not\to 0$, the series **diverges** by the n-th term test. (The partial sums oscillate without settling.)

Why Leibniz's test does not apply. Leibniz's test requires $|u_n| \to 0$ as one of its conditions. Here $|u_n| \to 1 \neq 0$, so the test is not applicable, and no conclusion of convergence may be drawn from the alternation of signs or from the fact that the denominators increase.

3 (B) — Convergence of a series in x [2]

Discuss the convergence of $\dfrac{1}{2\sqrt{1}} + \dfrac{x^2}{3\sqrt{2}} + \dfrac{x^4}{4\sqrt{3}} + \dfrac{x^6}{5\sqrt{4}} + \cdots$

The general term is

$$u_n = \frac{x^{2(n-1)}}{(n+1)\sqrt{n}}, \qquad n = 1, 2, 3, \ldots$$

Step 1 — the ratio test.

$$\left|\frac{u_{n+1}}{u_n}\right| = \frac{x^{2n}}{(n+2)\sqrt{n+1}} \cdot \frac{(n+1)\sqrt{n}}{x^{2(n-1)}} = x^2 \cdot \frac{n+1}{n+2} \cdot \sqrt{\frac{n}{n+1}} \longrightarrow x^2.$$

Hence the series converges for $x^2 < 1$, i.e. $|x| < 1$, and diverges for $x^2 > 1$, i.e. $|x| > 1$. The ratio test is *inconclusive* when $x^2 = 1$.

Step 2 — the boundary $x = \pm 1$. At either endpoint $x^2 = 1$ and the series becomes

$$\sum_{n=1}^{\infty} \frac{1}{(n+1)\sqrt{n}}.$$

Compare with $\sum n^{-3/2}$, which converges by the p-test ($p = \frac{3}{2} > 1$). By the limit comparison test,

$$\lim_{n\to\infty} \frac{1/\left[(n+1)\sqrt{n}\right]}{1/n^{3/2}} = \lim_{n\to\infty} \frac{n^{3/2}}{(n+1)\sqrt{n}} = \lim_{n\to\infty} \frac{n}{n+1} = 1,$$

a finite non-zero limit, so the boundary series **converges** as well.

Conclusion.

The series converges for $\lvert x\rvert \leq 1$ and diverges for $\lvert x\rvert > 1$.

Note that the interval is *closed*: both endpoints are included.

4 (A) — Linear and quadratic approximation of $\sqrt{4.05}$ [3]

Evaluate $\sqrt{4.05}$ *by linear and quadratic approximation of* $f(x) = \sqrt{x}$ *about* $a = 4$*; state the error in each case and comment on the improvement.*

Derivatives at the base point.

$$f(x) = x^{1/2}, \quad f'(x) = \tfrac{1}{2}x^{-1/2}, \quad f''(x) = -\tfrac{1}{4}x^{-3/2}; \qquad f(4) = 2, \quad f'(4) = \tfrac{1}{4}, \quad f''(4) = -\tfrac{1}{32}.$$

(i) Linear approximation.

$$L(x) = f(4) + f'(4)(x-4) = 2 + \tfrac{1}{4}(x-4), \qquad L(4.05) = 2 + \frac{0.05}{4} = \mathbf{2.01250000}.$$

(ii) Quadratic approximation.

$$Q(x) = f(4) + f'(4)(x-4) + \frac{f''(4)}{2!}(x-4)^2 = 2 + \tfrac{1}{4}(x-4) - \tfrac{1}{64}(x-4)^2,$$

$$Q(4.05) = 2.0125 - \frac{(0.05)^2}{64} = 2.0125 - 0.0000390625 = \mathbf{2.01246094}.$$

Comparison.

	Estimate	**True value**	**Magnitude of error**
Linear $L(4.05)$	2.01250000	2.01246118	3.882×10^{-5}
Quadratic $Q(4.05)$	2.01246094	2.01246118	2.422×10^{-7}

Comment. The quadratic term reduces the error by a factor of about **160** — from roughly 4×10^{-5} to 2×10^{-7}, which is two further correct decimal places. This is what the theory predicts: with $h = x - a = 0.05$, the linear error is of order h^2 and the quadratic error of order h^3, so the ratio of the two errors is of order $1/h = 20$ times a constant.

Consistency with the error bounds (not required for full marks).

$$\text{linear:} \quad \frac{|f''|_{\max}}{2!}\,h^2 = \frac{1/32}{2}(0.05)^2 = 3.906 \times 10^{-5},$$
$$\text{quadratic:} \quad \frac{|f'''|_{\max}}{3!}\,h^3 = \frac{3/256}{6}(0.05)^3 = 2.441 \times 10^{-7},$$

and in each case the actual error is just below its bound, as it must be.

4 (B) — Maclaurin series of $\sin x$ [2]

Obtain the Maclaurin series of $\sin x$ *and state the values of* x *for which it is valid.*

Derivatives at 0. With $f(x) = \sin x$,

$$f(0) = 0, \quad f'(0) = \cos 0 = 1, \quad f''(0) = -\sin 0 = 0, \quad f'''(0) = -\cos 0 = -1, \quad f^{(4)}(0) = \sin 0 = 0,$$

and the pattern 0, 1, 0, −1 repeats with period 4. Substituting into $\displaystyle\sum_{n\geq 0} \frac{f^{(n)}(0)}{n!}x^n$, the even-order terms vanish and

$$\boxed{\sin x \;=\; x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots \;=\; \sum_{n=0}^{\infty} \frac{(-1)^n\, x^{2n+1}}{(2n+1)!}}$$

Validity: for every $x \in \mathbb{R}$. The series represents $\sin x$ exactly when the Taylor remainder tends to 0. Every derivative of $\sin x$ is $\pm\sin x$ or $\pm\cos x$, so $\left|f^{(n+1)}(c)\right| \leq 1$ for all c and all n. By the Remainder Estimation Theorem,

$$|R_n(x)| \;=\; \left|\frac{f^{(n+1)}(c)}{(n+1)!}\,x^{\,n+1}\right| \;\leq\; \frac{|x|^{\,n+1}}{(n+1)!} \;\longrightarrow\; 0 \qquad \text{as } n \to \infty,$$

for every fixed x, since factorial growth outpaces any fixed power. Hence the equality holds on all of $\mathbb{R}$.